

Shreyas Thakur

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PROFESSIONAL SUMMARY

Chemical Engineering graduate from the Institute of Chemical Technology, Mumbai (2025) with hands-on industrial experience in continuous process operations, catalyst troubleshooting, and process safety at Epigral Ltd. Skilled in process simulation, HAZOP analysis, PFD/P&ID development, and equipment design. Seeking a Process Engineer / Chemical Engineer role to apply technical expertise in process optimization, plant operations, and scale-up to drive operational efficiency and safety excellence.

TECHNICAL SKILLS

Process Engineering: Process Design & Optimization, Mass & Energy Balances, Heat & Mass Transfer, Unit Operations, Fluid Mechanics, Reaction Engineering, Thermodynamics, Distillation, Absorption, Extraction
Safety & Compliance: HAZOP Analysis, Process Safety Management (PSM), Management of Change (MOC), Risk Assessment, Environmental Regulations, CPCB Standards, Effluent Treatment
Design & Documentation: PFD, P&ID, Equipment Sizing & Selection, Plant Layout, Piping Design, Process Flow Simulation
Software & Tools: Aspen HYSYS, MATLAB, Python, AutoCAD, MS Excel (Advanced), Power BI, MS Office Suite
Operations: DCS (Distributed Control System), Continuous Plant Operations, Batch Processing, SOPs, Root Cause Analysis
Languages: English (Fluent), Hindi (Fluent), Marathi (Native)

EDUCATION

Institute of Chemical Technology (ICT), Mumbai Mumbai, India
Bachelor of Technology – Chemical Engineering | CGPA: 6.6 2021 – 2025

- Relevant Coursework:** Chemical Reaction Engineering, Heat Transfer, Mass Transfer Operations, Fluid Mechanics, Process Dynamics & Control, Chemical Process Industries, Thermodynamics, Separation Processes, Plant Design & Economics
- HSC: 85.67% | SSC: 85.20%

INDUSTRIAL EXPERIENCE

In-Plant Trainee – Process Engineering Jun 2024 – Aug 2024
Epigral Ltd. (Formerly Meghmani Finechem) Dahej, Gujarat, India

- Performed **root cause analysis** on catalyst selectivity and deactivation issues in the **Hydrogen Peroxide** manufacturing unit, contributing to recommendations that improved catalyst lifecycle monitoring
- Investigated recurring **burner failures** in HCl synthesis units; documented failure modes and proposed corrective actions aligned with **process safety management (PSM)** protocols
- Conducted a **moisture-reduction study** in the Epichlorohydrin (ECH) purification section, identifying process bottlenecks and recommending optimization strategies for improved product purity
- Operated and monitored **continuous chemical processes** via **DCS (Distributed Control System)**, gaining real-time exposure to process parameters, alarms, and control loops across multiple production units
- Completed training in **industrial safety protocols**, including HAZOP procedures, permit-to-work systems, and **Management of Change (MOC)** documentation for plant modifications
- Gained exposure to chlor-alkali processes, caustic soda, hydrogen peroxide, and ECH production – covering reactor operations, distillation columns, heat exchangers, and effluent treatment systems

ENGINEERING PROJECTS

Complete Plant Design – Ambrettolide Manufacturing (25 TPA) | Final Year Capstone Sep 2024 – Apr 2025

- Designed an **end-to-end chemical manufacturing plant** for 25 TPA Ambrettolide (macrocyclic musk compound) from concept to detailed engineering
- Evaluated 3+ synthesis routes using techno-economic analysis; selected optimal process based on yield, safety, raw material cost, and scalability

- Performed rigorous **material and energy balance** calculations across all unit operations including reactors, distillation columns, condensers, and heat exchangers
- Sized and designed key equipment: **CSTR reactor, packed distillation column, shell-and-tube heat exchangers, storage tanks**, and pumps with mechanical design specifications
- Developed comprehensive **PFD and P&ID** drawings; conducted detailed **HAZOP analysis** identifying 15+ deviation scenarios with safeguards
- Prepared complete **project cost estimation** (CAPEX & OPEX), breakeven analysis, ROI calculation, site layout, and **effluent treatment scheme** meeting environmental norms

Impact of Climate Change on Water Quality & Treatment Strategies | *Technical Seminar* Nov 2024 – Jan 2025

- Authored a comprehensive technical review analyzing climate change impacts on surface water, groundwater, and coastal water systems with reference to **CPCB standards**
- Evaluated advanced water treatment technologies including **membrane separation (UF/NF/RO)**, **Advanced Oxidation Processes (AOPs)**, and biological treatment for emerging contaminants
- Benchmarked conventional vs. advanced treatment processes on efficiency, cost, and scalability for industrial and municipal applications

CERTIFICATIONS & TRAINING

Petroleum Refinery Engineering | *Indian Institute of Chemical Engineers (IChE)* Feb 2022 – Apr 2022

- Completed industry-focused certification covering crude oil processing, catalytic cracking, reforming, hydroprocessing, and refinery unit operations

Industrial Environmental Pollution Management | *Terra-Green Technologies Pvt. Ltd.* Jun 2022 – Jul 2022

- Trained in industrial pollution control techniques, air/water/solid waste treatment, environmental impact assessment (EIA), and regulatory compliance

Industrial Internet of Things (IIoT) | *NPTEL – Top 5% Nationwide* 2023

- Secured **Top 5% rank** nationally; gained knowledge in sensor networks, SCADA, smart manufacturing, and Industry 4.0 applications in chemical process industries

LEADERSHIP & ACHIEVEMENTS

Creative Head | *Bombay Technologist – ICT Mumbai's Official Technical Magazine* Sep 2023 – Feb 2024

- Led a team of 10+ members for content strategy, design, and publication of ICT Mumbai's flagship technical magazine

Competitions & Recognition

- **Top 5%** nationally in NPTEL Industrial Internet of Things certification (2023)
- **2nd Place** – “The Entrepreneur's Blueprint” business plan competition at Vortex 2022, ICT Mumbai